Session 5 — Advanced Probability & Statistics

What today is about

This session fuses the probability of Sessions 1–2 with the linear algebra of Sessions 3–4 into the tools that models actually use. It is a recap and an elevation for students who already have some mathematics.

- *Bayesian inference* over a whole parameter.
- The *multivariate Gaussian* — where the covariance matrix meets PCA.
- *Maximum likelihood*: the one principle behind most model fitting.
- The *GLM*, *model comparison*, *information theory*, and a taste of *sampling*.

Bayes as a framework, not just a formula

In Session 1, Bayes updated the probability of a single hypothesis. As a *framework*, it updates a whole distribution over a continuous parameter.

Key idea

$$p(\theta \mid D) = \frac{p(D \mid \theta)\, p(\theta)}{p(D)} \propto \underbrace{p(D \mid \theta)}_{\text{likelihood}}\, \underbrace{p(\theta)}_{\text{prior}}.$$

We start with a prior distribution over θ, weight it by how well each θ predicts the data, and renormalise to get the posterior.

The answer is a distribution, not a point

Intuition

The Bayesian answer is never a single number — it is a whole distribution that carries its own uncertainty. A sharp posterior means the data pinned θ down; a broad one means we are still unsure.

Two common summaries: the **posterior mean** (the average belief), and the **MAP estimate** (the single most probable value, the peak of the posterior).

Bayesian estimation — worked example

Beta–Binomial

We want the probability θ that a participant detects a faint stimulus. Put a uniform prior $\theta \sim \mathrm{Beta}(1, 1)$ and observe 8 detections in 10 trials.

For a $\mathrm{Beta}(a, b)$ prior and h successes in $h + t$ trials, the posterior is $\mathrm{Beta}(a + h,\ b + t)$. Here it is $\mathrm{Beta}(9, 3)$, with posterior mean $\frac{9}{12} = 0.75$ and MAP $\frac{8}{10} = 0.80$. The prior gently pulls the estimate away from the extreme, and the posterior's width honestly reports the remaining uncertainty.

Two facts that make Bayes practical

Useful fact

Sequential updating: today's posterior is tomorrow's prior — processing data all at once or one point at a time gives the *same* posterior, so beliefs can be updated online.

Credible interval: a range holding 95% of the posterior. Unlike a confidence interval, it *does* allow the statement "there is a 95% probability θ is in here".

Bayes as a theory of cognition

Link to cognitive science

Bayesian inference is the dominant formal language of contemporary cognitive modeling: perception as inference, cue combination, learning, and the "Bayesian brain" all cast cognition as computing posteriors. The prior encodes expectations, the likelihood encodes sensory evidence, and behaviour reflects the posterior.

The prior is a modeling choice

Common pitfall

The prior is a deliberate choice, not a nuisance to hide. With little data it matters; with abundant data the likelihood dominates and the posterior becomes insensitive to any reasonable prior.

Good practice is to state priors explicitly and check that conclusions are robust to sensible alternatives.

Practice — Bayesian updating

Practice — try it

Start from a uniform $\mathrm{Beta}(1, 1)$ prior on a detection probability θ.

1. After 3 hits in 3 trials, what is the posterior?
2. What is its posterior mean?
3. You then see 1 hit in the next 2 trials — update again.

Several variables at once

Real data have many variables that must be modelled *jointly*.

Definition

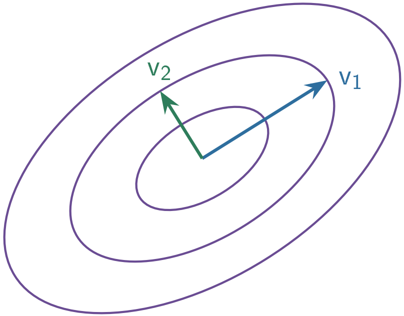
The **joint** distribution $p(x, y)$ describes variables together; the **marginal** $p(x) = \sum_y p(x, y)$ describes one alone; the **conditional** $p(x \mid y)$ describes one given another. They are linked by $p(x, y) = p(x \mid y)\, p(y)$.

The multivariate Gaussian

Key idea

$\mathcal{N}(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ generalises the bell curve to many dimensions. It is defined by a **mean vector** $\boldsymbol{\mu}$ (the centre) and a **covariance matrix** $\boldsymbol{\Sigma}$ (the shape). Its contours of equal probability are ellipses whose axes are the *eigenvectors* of $\boldsymbol{\Sigma}$ and whose widths are set by its *eigenvalues* — exactly the PCA picture.

The covariance ellipse



Equal-probability contours of a 2-D Gaussian. The ellipse axes are the eigenvectors of the covariance matrix; their lengths grow with the eigenvalues. This is the same picture as PCA.

The object that needs both toolkits

Intuition

The covariance matrix is where statistics and linear algebra become the same subject. Statistics supplies its entries (variances and covariances); linear algebra reads its shape (eigenvectors and eigenvalues give the orientation and spread of the data ellipse).

This is the single most important unification in the course: “covariance”, “eigenvector”, and “principal component” turn out to be three views of one thing.

Why the Gaussian is so convenient

Useful fact

- **Closed** under marginals and conditionals: subsets and conditionals of a Gaussian are again Gaussian.
- For Gaussians, *uncorrelated* implies *independent* — so a diagonal covariance matrix means fully independent components.
- **Linear maps** keep it Gaussian: $Ax + b \sim \mathcal{N}(A\boldsymbol{\mu} + b,\ A\Sigma A^\top)$, which is exactly what PCA's rotation uses.

The multivariate Gaussian in practice

Link to cognitive science

Multivariate Gaussians underlie models of neural noise correlations, Gaussian classifiers and linear discriminant analysis, Kalman filters for tracking and motor control, and the latent-variable models (factor analysis, probabilistic PCA) used to describe population activity.

Whenever someone says “noise is correlated across neurons”, a covariance matrix is doing the work.

One principle for fitting almost any model

Definition

The **maximum likelihood estimate** is the parameter value that makes the observed data most probable:

$$\hat{\theta}_{\mathrm{MLE}} = \arg\max_{\theta} \; p(D \mid \theta) = \arg\max_{\theta} \sum_i \log p(x_i \mid \theta),$$

where we maximise the *log*-likelihood (a sum, easier than a product).

Likelihood reverses the usual question

Intuition

A distribution asks: "given the parameters, how probable is each possible dataset?" Likelihood fixes the data we actually saw and asks the reverse: "which parameters would have made *this* most plausible?" Maximum likelihood simply picks that best-explaining parameter.

Taking the log turns the product over independent data points into a sum, which is what we actually optimise — and it does not move the maximum.

Maximum likelihood unifies what you know

Key idea

For Gaussian data, the MLE of the mean is just the *sample mean*. Fitting a regression line by *least squares* is exactly maximum likelihood under Gaussian noise. So "minimise squared error" and "maximise likelihood" are the same procedure in disguise — and in machine learning, the negative log-likelihood is the "loss" we minimise.

Deriving the obvious estimate the formal way

Example

A participant succeeds on k of n independent trials. Find the MLE of the success probability p.

The log-likelihood is $\ell(p) = k \ln p + (n-k)\ln(1-p)$. Setting the derivative to zero:

$$\frac{k}{p} - \frac{n-k}{1-p} = 0 \implies \hat{p} = \frac{k}{n}.$$

Maximum likelihood recovers the natural estimate — the observed success rate — and the same machinery works when the answer is *not* obvious.

Practice — likelihood

Practice — try it

1. A neuron responds on 12 of 40 trials. What is the MLE of its response probability?
2. Two models: A says $p = 0.5$, B says $p = 0.7$. You see 7 heads in 10 flips. Which assigns higher likelihood?
3. Why maximise the *log*-likelihood rather than the likelihood itself?

Extending regression to other data types

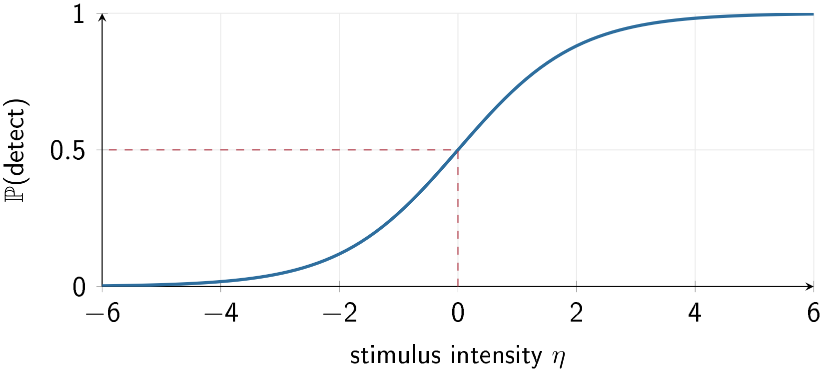
Regression assumed a straight line and Gaussian noise. Many outcomes are binary (correct/incorrect) or counts (spikes). The GLM extends regression to these while keeping its linear core.

Definition

A **Generalized Linear Model** keeps a linear predictor $\eta = \beta_0 + \beta_1 x_1 + \dots$, but passes it through a **link function**. For binary outcomes, *logistic regression* uses the logistic (sigmoid) link:

$$\mathbb{P}(y = 1 \mid x) = \sigma(\eta) = \frac{1}{1 + e^{-\eta}}.$$

The psychometric function



The sigmoid squashes any real number into a probability in $[0, 1]$, producing the familiar S-shaped psychometric function.

What the link function does, and what coefficients mean

Intuition

The link is a translator. The linear part η can be any number, but a probability must live in $[0, 1]$. The sigmoid squashes the whole real line into that range, giving the smooth rise from 0 to 1 as stimulus intensity increases.

Coefficients are log-odds

$\ln \dfrac{\mathbb{P}(y = 1)}{\mathbb{P}(y = 0)} = \beta_0 + \beta_1 x_1 + \dots$, so each β_j is the change in log-odds per unit of x_j, and e^{β_j} is the **odds ratio**.

The GLM in cognitive science

Link to cognitive science

Logistic regression *is* the psychometric function relating stimulus to detection probability. Poisson regression models spike counts as a function of stimuli (the linear–nonlinear–Poisson encoding model). The same framework yields decoding models that read stimuli off neural activity.

Master the GLM and a large family of analysis methods becomes one familiar object.

Practice — the GLM

Practice — try it

A logistic model of "detected" has $\eta = \beta_0 + \beta_1 x$ with $\beta_0 = 0$.

1. At $\eta = 0$, what is the detection probability?
2. As x grows large, what does the probability approach?
3. If $\beta_1 = \ln 2$, by what factor do the odds of detection change per unit of x?

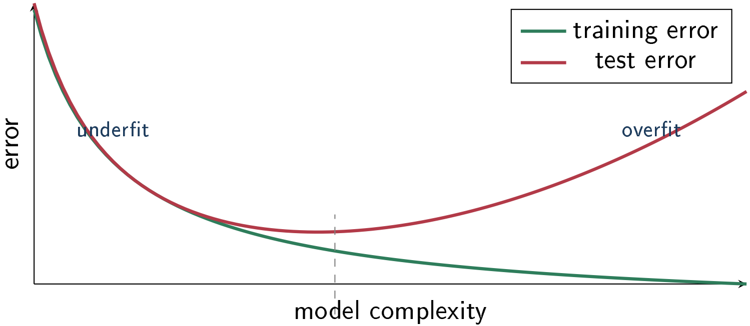
The most transferable skill in the course

A model that fits the training data perfectly may predict new data terribly. Guarding against this is what separates fitting from understanding.

The bias–variance trade-off

Too simple a model *underfits* (high bias — it misses real structure). Too complex a model *overfits* (high variance — it chases noise). Good modeling finds the middle: flexible enough to capture the signal, not so flexible that it memorises the noise.

Underfitting, overfitting, and the sweet spot



Training error always falls with complexity, but test error turns back up when the model starts fitting noise. The best complexity is where test error bottoms out.

Two practical tools

Key idea

Cross-validation: hold out part of the data, fit on the rest, and measure prediction error on the held-out part. This directly estimates generalisation — the gold standard.

Information criteria: score a model by fit *penalised* for complexity, $\mathrm{AIC} = 2k - 2\ln\hat{L}$ and $\mathrm{BIC} = k\ln n - 2\ln\hat{L}$. Lower is better.

AIC and BIC can disagree

Example

Two models fit to $n = 100$ points. Model 1: $k = 2$, $\ln \hat{L} = -50$. Model 2: $k = 4$, $\ln \hat{L} = -47$ (a little better, as a more flexible model must be).

$$\text{AIC}_1 = 104, \quad \text{AIC}_2 = 102 \;\Rightarrow\; \text{AIC picks Model 2,}$$

$$\text{BIC}_1 \approx 109.2, \quad \text{BIC}_2 \approx 112.4 \;\Rightarrow\; \text{BIC picks Model 1.}$$

BIC's heavier penalty ($\ln n \approx 4.6$ per parameter vs. AIC's 2) favours the simpler model. When criteria disagree, cross-validation is the tie-breaker.

The one mistake to never make

Common pitfall

Never judge a model only by how well it fits the data it was trained on — more parameters *always* fit training data better. The honest question is always out-of-sample prediction.

Practice — try it

1. Adding parameters lowers training error. Does that make the model better?
2. Two models with equal test error — which do you keep?
3. Which penalises complexity more for $n > 7$, AIC or BIC?

Measuring uncertainty and surprise

Definition

Entropy $H(X) = -\sum_x p(x)\log_2 p(x)$ is the average uncertainty (in bits). **KL divergence** $D_{\mathrm{KL}}(p\|q) \geq 0$ measures how different q is from a reference p. **Mutual information** $I(X;Y)$ measures how much knowing Y reduces uncertainty about X.

Entropy — intuition and example

Intuition

Entropy is "how surprised do I expect to be?" A fair coin has entropy 1 bit (maximally uncertain); a two-headed coin has entropy 0 (no surprise). KL divergence is the extra cost of describing reality with the wrong distribution; mutual information is the information-theoretic cousin of correlation, but sensitive to *any* dependence.

Example

A coin with $p = 0.25$: $H = -0.25 \log_2 0.25 - 0.75 \log_2 0.75 \approx 0.81$ bits — less than the 1 bit of a fair coin, because the bias makes it more predictable.

A few facts, and why they matter for ML

Useful fact

The uniform distribution maximises entropy ($\log_2 m$ bits). Cross-entropy splits as $H(p,q) = H(p) + D_{\mathrm{KL}}(p\|q)$ — which is why minimising cross-entropy loss trains a classifier toward the true distribution. And $I(X;Y) = H(X) - H(X \mid Y) \geq 0$, equal to 0 exactly when X and Y are independent.

Link to cognitive science

Information theory quantifies how much a neuron tells you about a stimulus (mutual information), underlies the efficient-coding hypothesis, and supplies the loss functions of machine learning.

Practice — information

Practice — try it

1. What is the entropy of a fair six-sided die?
2. Which has more entropy: a fair coin, or a coin with $p = 0.9$?
3. If knowing Y makes X certain, what is $I(X;Y)$ in terms of $H(X)$?

How Bayesian models are actually computed

Definition

A **Markov chain** is a sequence of states in which the next state depends only on the current one (the *Markov property*). Its dynamics are captured by a **transition matrix** — so propagating the state forward is repeated matrix multiplication, a nice callback to linear algebra.

Sampling: replacing integrals with histograms

Intuition

Posteriors in realistic models are too complicated to write down or integrate. So instead of computing them, we *sample*. **Markov chain Monte Carlo** builds a chain that wanders through parameter space, spending time in each region in proportion to its posterior probability. The collected samples then approximate the posterior — histograms stand in for integrals.

Link to cognitive science

Sampling has even been proposed as a model of cognition itself: perhaps the brain represents uncertainty by drawing samples. Markov models also describe sequential behaviour and spike-train dynamics.

Session 5 — and the whole course

- **Bayes** updates a full distribution (prior $\rightarrow$ posterior; mean or MAP).
- The **multivariate Gaussian** unites both subjects through the covariance matrix, whose eigenstructure is PCA.
- **Maximum likelihood** is the fitting principle; least squares is a special case.
- The **GLM** extends regression; **model comparison** fights overfitting; **information theory** measures uncertainty; **MCMC** computes posteriors.

You now hold the mathematical vocabulary of modern cognitive modeling, computational neuroscience, machine learning, and data analysis.